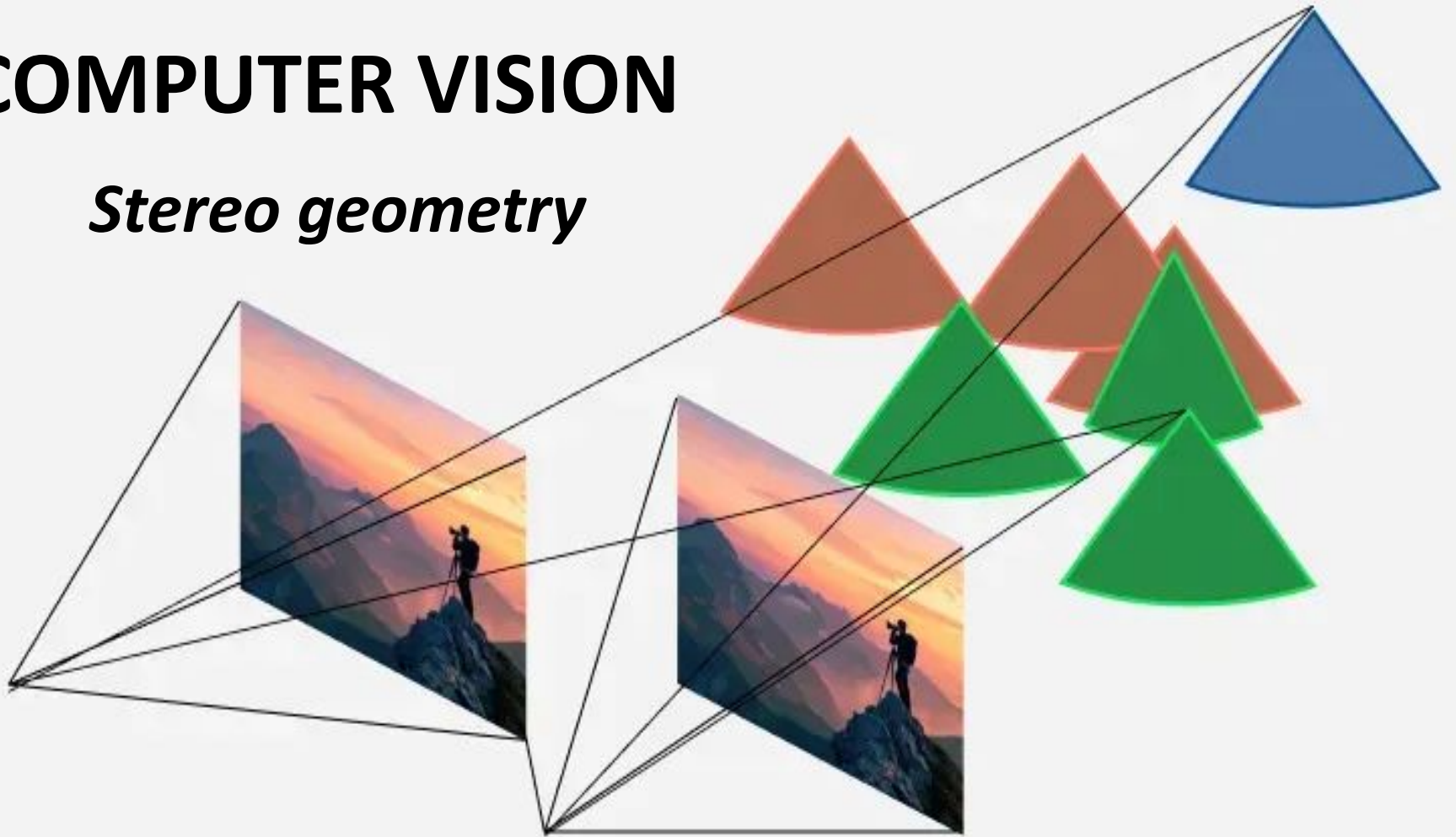


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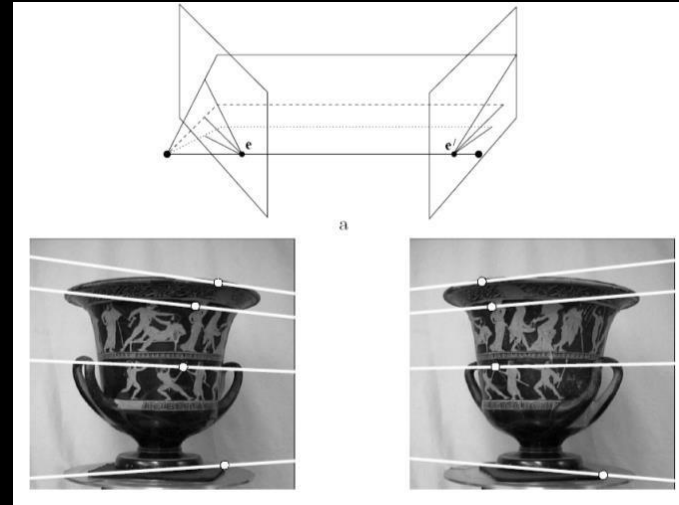
Stereo geometry



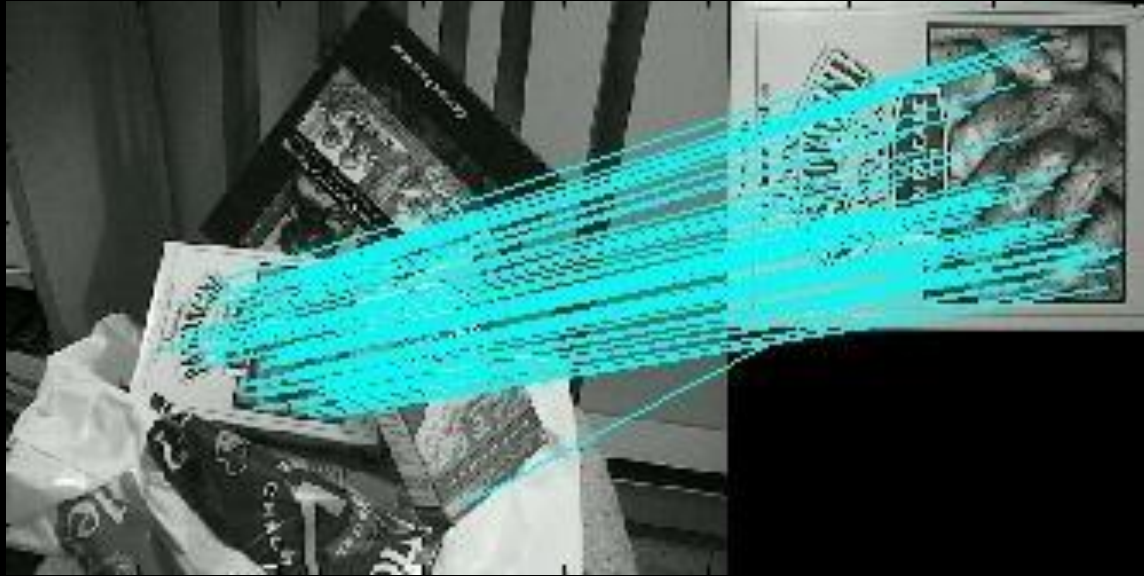
Camera geometry vs Scene geometry

Stereo: Multiple views of the same image

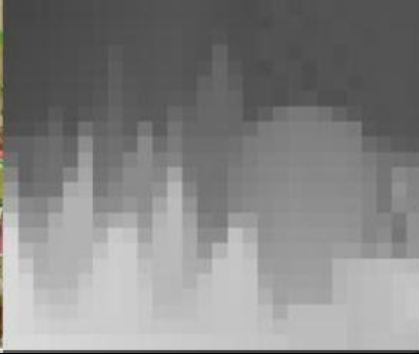
View of same
object in different
ways



Stereo: A Special case of Multiple views

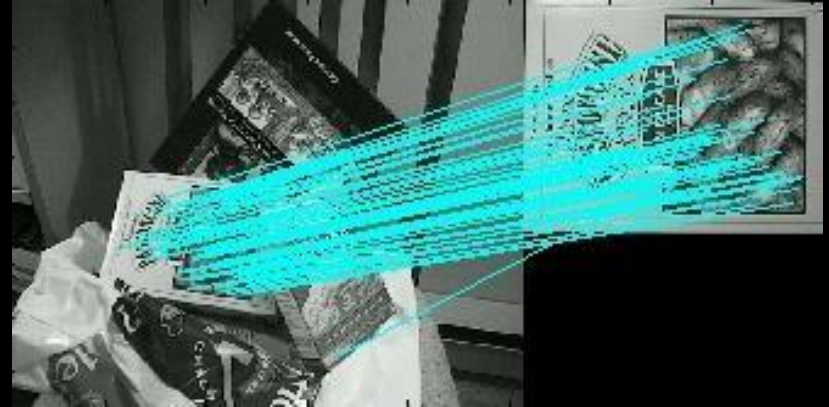


Stereo: A Special case of Multiple views



Half of the
stereo pair

crude depth map
brighter => closer



Why multiple views? 3D => 2D information loss

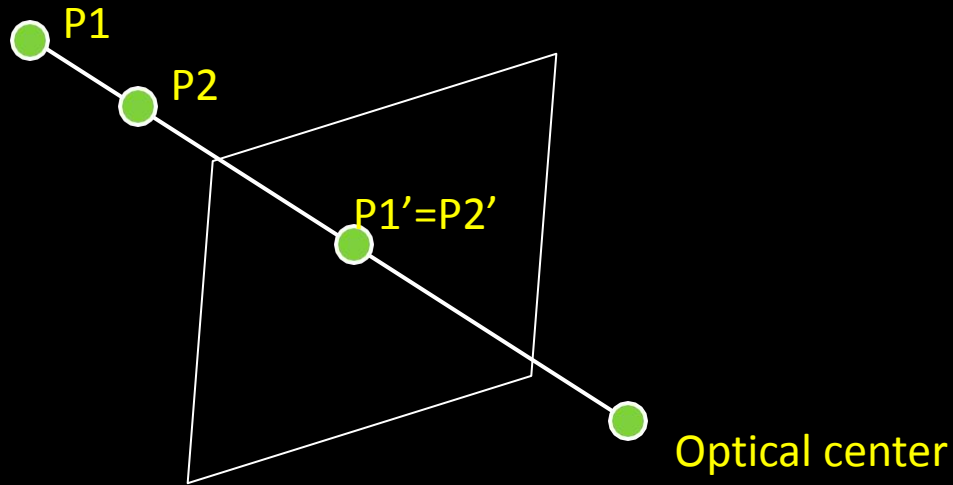
- Structure and depth are inherently ambiguous from single views.



Location of camera has an effect on what we see.

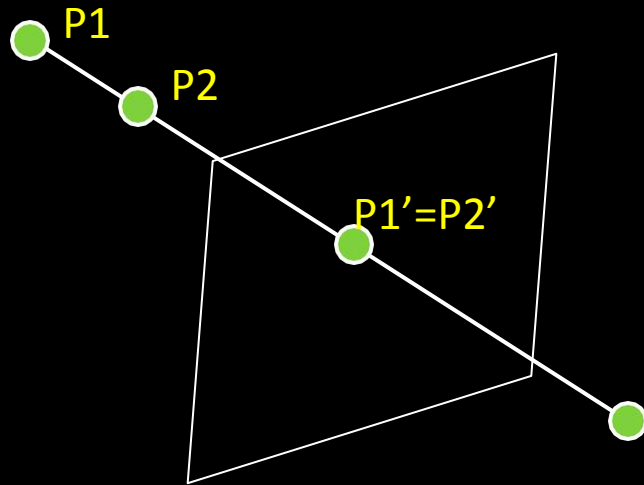
Why multiple views?

- Structure and depth are inherently ambiguous from single views.



Why multiple views?

- Structure and depth are inherently ambiguous from single views.

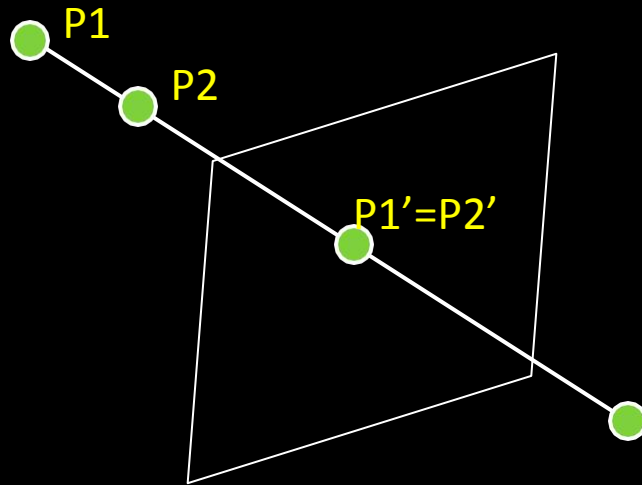


Problem =>
How to figure out
exactly where P1'
and P2' are?

Optical center

Why multiple views?

- Structure and depth are inherently ambiguous from single views.



Look at using
different cameras

Optical center

Perspective effects



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S. Seitz

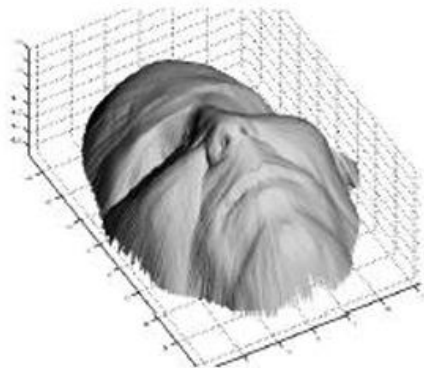
Shading



a)

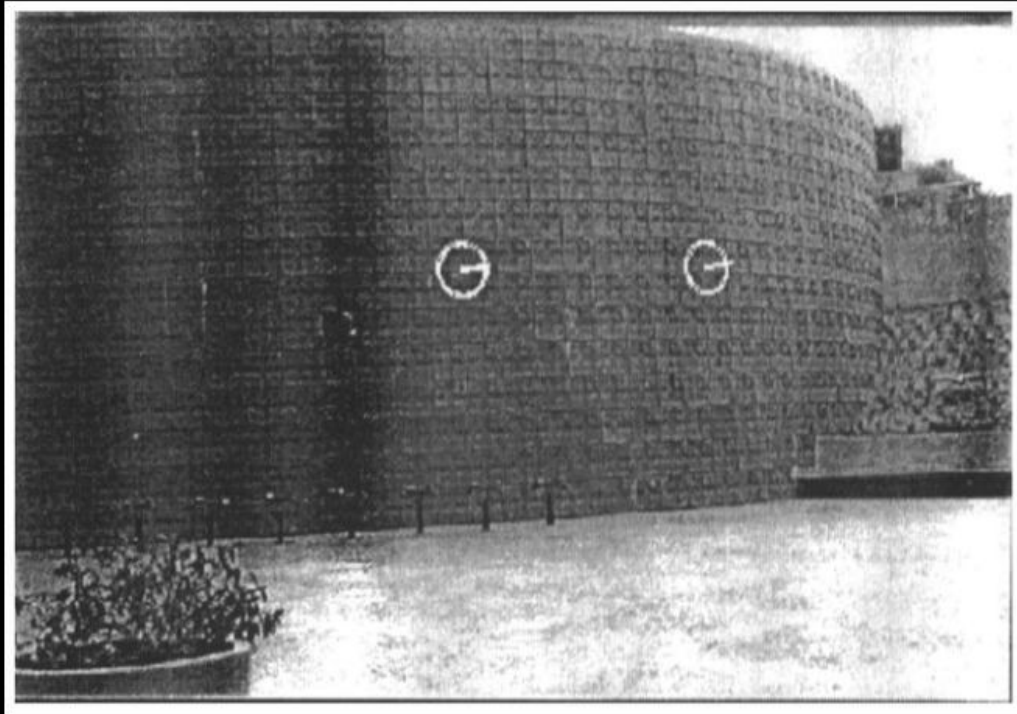


b)



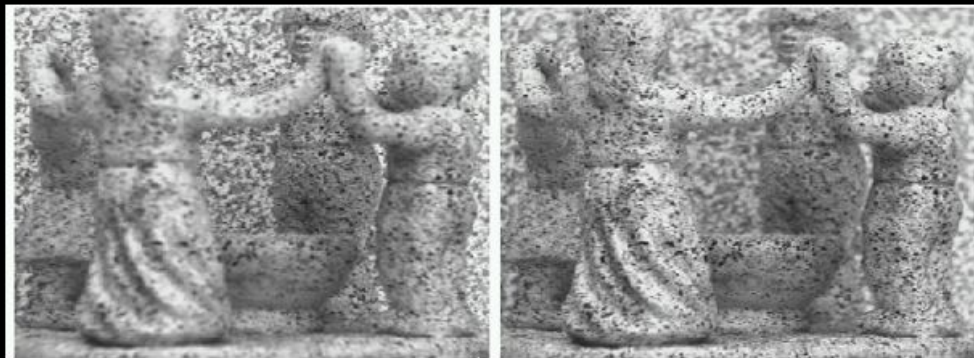
c)

Texture

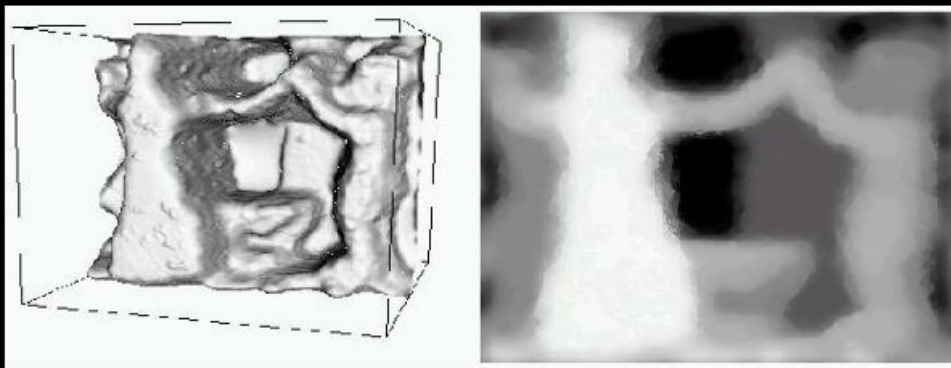


A.M. Loh. The recovery of 3-D structure using visual texture patterns.

Focus/defocus



Images from same
point of view, different
camera parameters



3d shape / depth
estimates

Figures from H. Jin and P. Favaro, 2002

But we (and lots of creatures) have two eyes!

Stereo:

- The image from one eye is a little different than the image from the other eye.

Stereo photography and stereo viewers

Take two pictures of the same subject from two slightly different viewpoints and display so that each eye sees only one of the images.

Invented by Sir Charles Wheatstone
1838

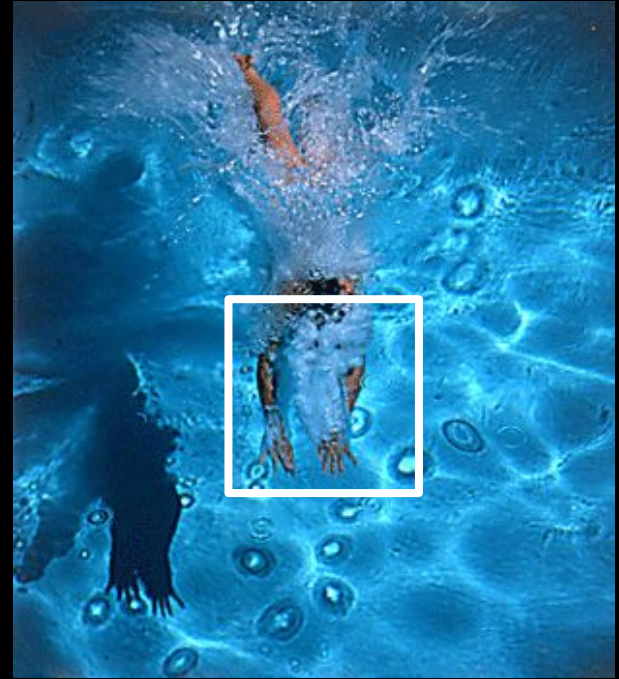


The Basic Idea: Two slightly different images



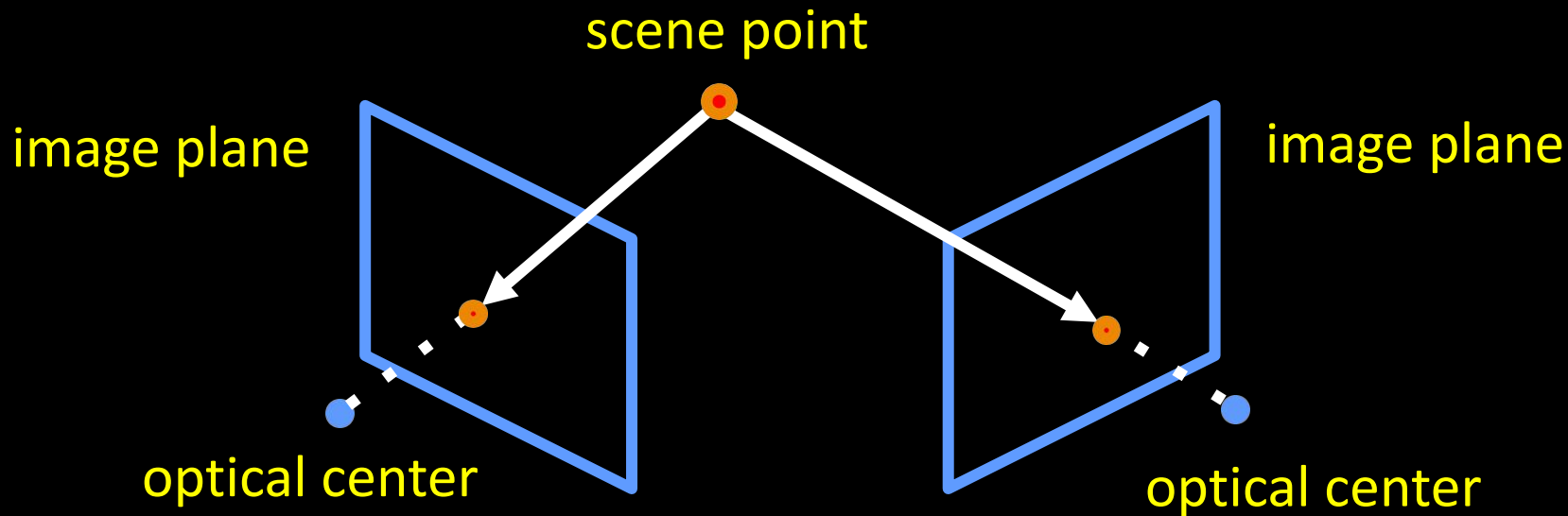
Aligning left image and right image

The Basic Idea: Two slightly different images



From two different views, by observing how they move we can obtain a sense of depth information

Basic stereo geometry



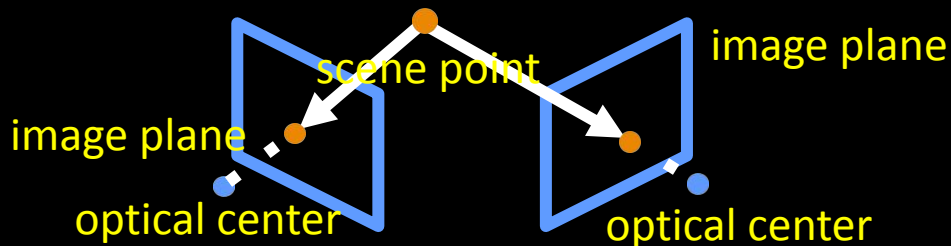
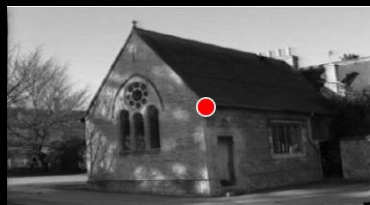
which two points

Estimating depth with stereo

Stereo: shape from “motion” between two views

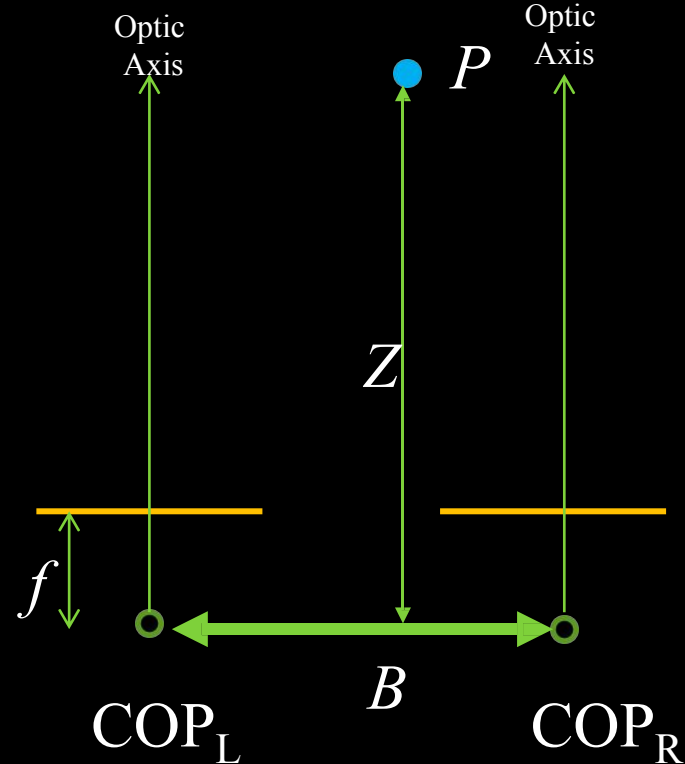
We'll need to consider:

- Info on camera pose (“calibration”)
- Image point correspondences



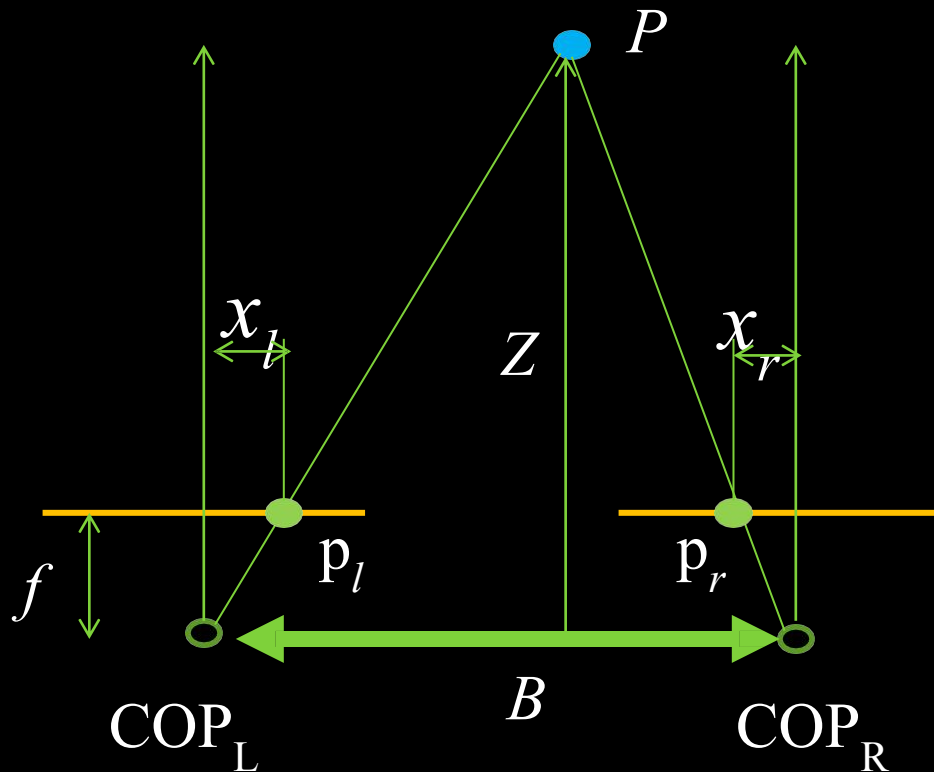
Geometry for a simple stereo system

- First, assuming parallel optical axes, known camera parameters (i.e., calibrated cameras)
- Figure is looking down on the cameras and image planes
- Baseline B , focal length f
- Point P is distance Z in camera coordinate systems



Geometry for a simple stereo system

- Point P projects into left and right images.
- Distance is positive in left image, and negative in right

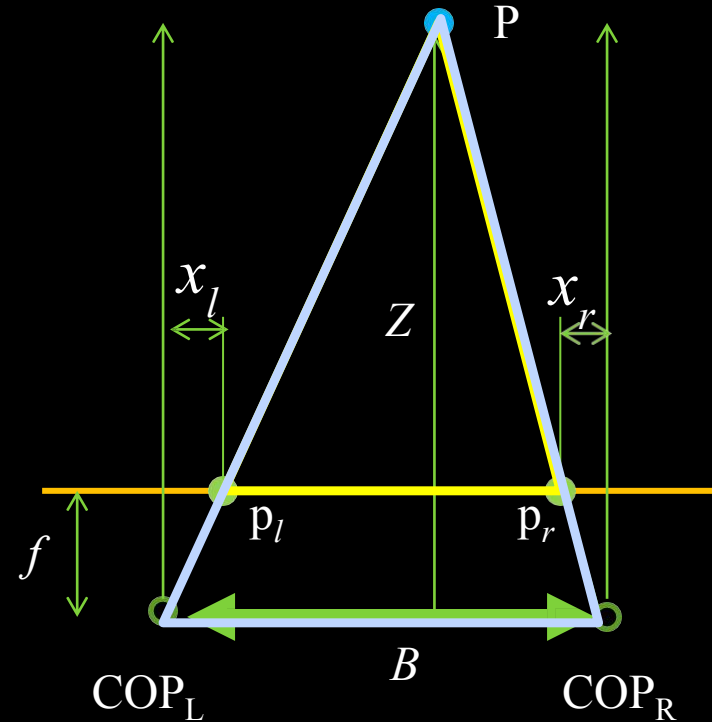


Geometry for a simple stereo system

- What is the expression for Z ?
- Similar triangles (p_l, P, p_r) and (C_L, P, C_R) :

$$\frac{B - x_l + x_r}{Z - f} = \frac{B}{Z}$$

$$Z = f \frac{B}{x_l - x_r}$$



Disparity

Geometry for a simple stereo system

- What happen when disparity is large/ small/ zero

$$\frac{B - x_l + x_r}{Z - f} = \frac{B}{Z}$$

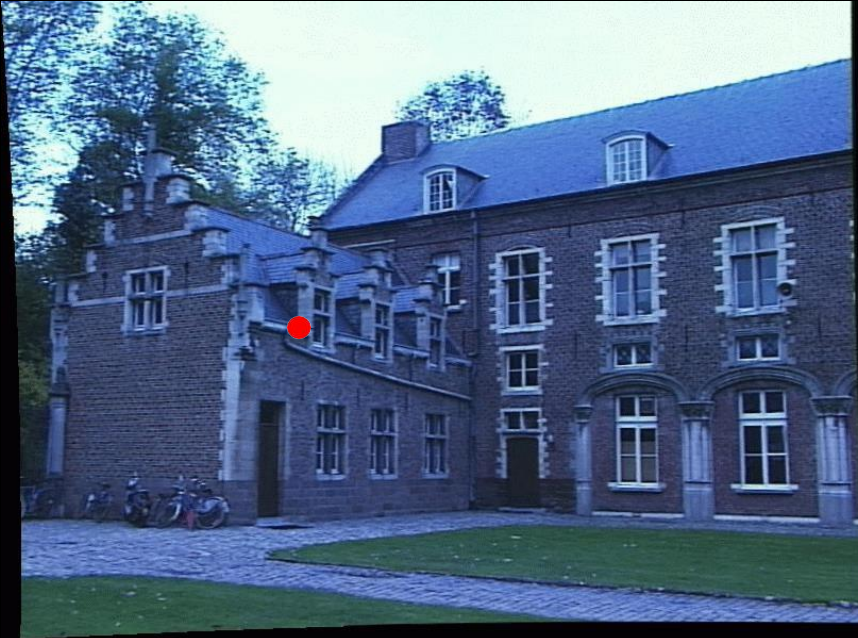
$$Z = f \frac{B}{x_l - x_r}$$

Disparity

Depth from disparity

image $I(x,y)$

image $I'(x,y)$



left



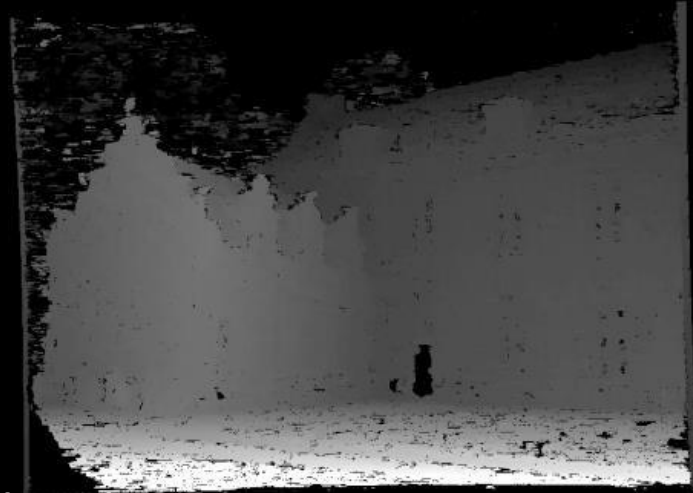
right

Depth from disparity

image $I(x,y)$



Disparity map $D(x,y)$



disparity inversely proportional to depth

Brighter values => Closer Z value

Very dark => small values, very further away

Depth from disparity

$$(x', y') = (x + D(x, y), y)$$

image $I'(x', y')$

If we have calibrated cameras & if we know the match we can compute the depth



Depth from disparity

$$(x', y') = (x + D(x, y), y)$$

image $I'(x', y')$

next =>

Finding disparity and
figuring out which point in
one matches the other



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Epipolar geometry

